

read mesh and make Graphics primitives

```
fn = "~/Documents/NACA0012/NACA0012.msh";

stream1 = OpenRead[fn];
{n1, n2, n3} = Read[stream1, {Number, Number, Number}]
{3034, 5876, 192}

frm = Table[{Real, Real, Number}, {n1}];

p = Read[stream1, frm];

p1 = Table[{p[[i, 1]], p[[i, 2]]}, {i, Length[p]}];

frm = Table[{Number, Number, Number, Number}, {n2}];

tr = Read[stream1, frm];

tr1 = Table[{tr[[i, 1]], tr[[i, 2]], tr[[i, 3]], tr[[i, 4]]}, {i, Length[tr]}];

frm = Table[{Number, Number, Number}, {n3}];

vt = Read[stream1, frm];

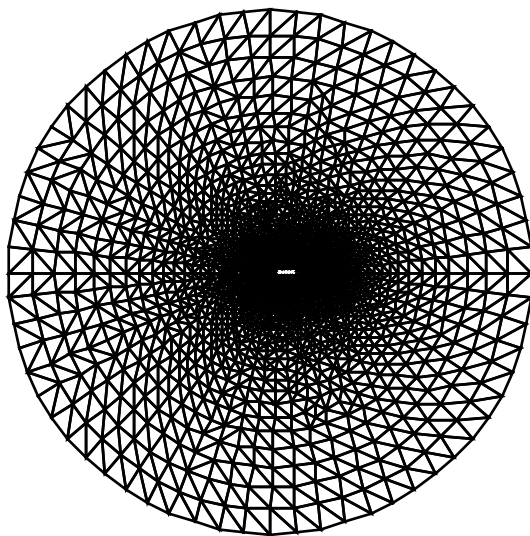
Close[stream1];

p2 = Map[p1[[#]] &, tr1];

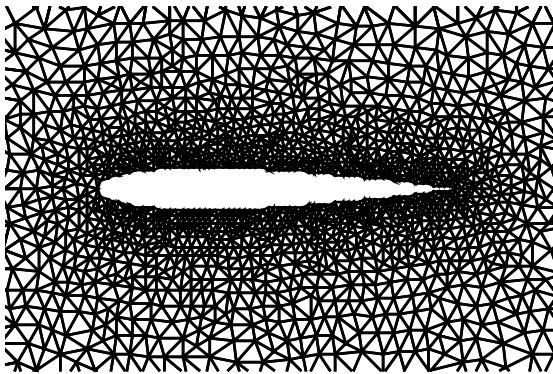
p2[[1]]
{{0.942677, -0.0245889}, {0.926947, -0.0261184},
 {0.932553, -0.0433022}, {0.942677, -0.0245889}}

lp1 = Map[Line, p2];

Show[Graphics[lp1], AspectRatio -> 1, PlotRange -> All];
```



```
Show[Graphics[lp1], AspectRatio → 1 / 1.5, PlotRange → {{-.25, 1.25}, {-.5, .5}}];
```



Read values of the solution

```
fn = "~/Documents/NACA0012/NACA0012.out";

stream1 = OpenRead[fn];

n4 = Read[stream1, Number]

3034

frm = Table[Real, {n4}];

r = Read[stream1, frm];

Close[stream1];

Dimensions[r]

{3034}
```

Normalization of values of solution

```
m1 = Min[r];

r = r - m1;

m1 = Max[r];

r = r / m1;

r2 = Table[(r[[tr[[i, 1]]] + r[[tr[[i, 2]]] + r[[tr[[i, 3]]] / 3.0, {i, Length[tr]]];
```

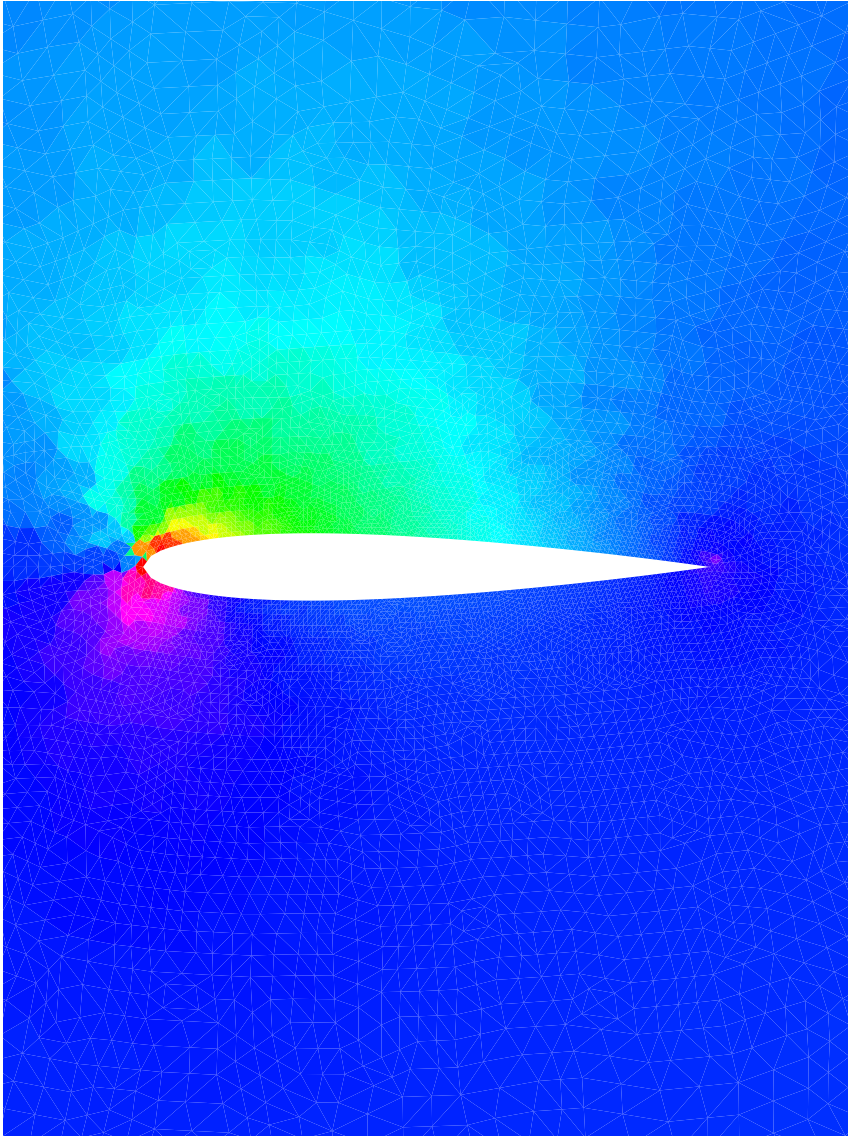
Make Colored polygon

```
p3 = Table[{(p1[[tr[[i, 1]]] + p1[[tr[[i, 2]]] * 0.5, (p1[[tr[[i, 2]]] + p1[[tr[[i, 3]]] * 0.5,
(p1[[tr[[i, 3]]] + p1[[tr[[i, 1]]] * 0.5), {i, Length[tr]]];

plg = Table[{Hue[r[[tr[[i, 1]]]], Polygon[{p2[[i, 1]], p3[[i, 1]], p3[[i, 3]]}],
{Hue[r2[[i]], Polygon[{p3[[i, 1]], p3[[i, 2]], p3[[i, 3]]}], {Hue[r[[tr[[i, 2]]]],
Polygon[{p2[[i, 2]], p3[[i, 2]], p3[[i, 1]]}], {Hue[r[[tr[[i, 3]]]],
Polygon[{p2[[i, 3]], p3[[i, 3]], p3[[i, 2]]}]}, {i, Length[tr]]];
```

Draw the result

```
Show[Graphics[plg], PlotRange → {{-.25, 1.25}, {-1, 1}}, AspectRatio → 2 / 1.5];
```



```
Show[Graphics[plg], PlotRange -> All, AspectRatio -> 1];
```

